

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I C.ELIYAZAR/192311162 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

C. Eliyazar

Signature of the student:

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Solution :

The Caesar Cipher (Shift Cipher)

The scenario describes a classical substitution cipher known as the **Caesar Cipher**, one of the earliest and simplest encryption techniques. Each letter of the plaintext is replaced by a letter a fixed number of positions ("shift" or "key") further along in the alphabet.

Encryption formula:

$$C = (P + k) \bmod 26$$

Decryption formula:

$$P = (C - k) \bmod 26$$

Where:

- P = numeric position of plaintext letter (A=0, B=1, ..., Z=25)
- C = numeric position of ciphertext letter
- k = shift key (here, k = 4)
- Spaces and non-alphabetic characters are typically left unchanged (they carry no cryptographic value in this scheme).

a. Encrypting "MEET AT NOON" with a shift of 4

Let's map each letter to its position, add 4, and wrap around (mod 26) if it exceeds 'Z':

Plaintext	Position (P)	P + 4	Mod 26	Ciphertext
M	12	16	16	Q
E	4	8	8	I
E	4	8	8	I

Plaintext	Position (P)	P + 4	Mod 26	Ciphertext
T	19	23	23	X
A	0	4	4	E
T	19	23	23	X
N	13	17	17	R
O	14	18	18	S
O	14	18	18	S
N	13	17	17	R

Encrypted Message: QIIX EX RSSR

(Spaces are preserved as word separators; only alphabetic characters are shifted.)

b. How an authorized member retrieves the original message

An authorized member who knows the encoding rule (i.e., that a **shift of 4** was used) can recover the plaintext through the **reverse operation — decryption**:

1. **Know the shared secret key:** In classical symmetric-key cryptography, both sender and receiver must agree on the key (here, $k = 4$) in advance through a secure channel. This shared secret is what makes decryption possible only for authorized members.
2. **Apply the inverse shift:** For each letter in the ciphertext, the member subtracts the key value (4) from its position, instead of adding it:

$$P = (C - k) \bmod 26$$

3. **Reconstruct the message letter by letter:**

Ciphertext	Position (C)	C - 4	Mod 26	Plaintext
Q	16	12	12	M
I	8	4	4	E
I	8	4	4	E
X	23	19	19	T
E	4	0	0	A
X	23	19	19	T
R	17	13	13	N
S	18	14	14	O
S	18	14	14	O
R	17	13	13	N

Decrypted Message : MEET AT NOON

(b) Explain how an authorized member retrieves the original message

An authorized member already knows that the sender used a **Caesar Cipher with a shift of 4**.

To recover the original message, they perform **decryption**, which means shifting every encrypted letter **backward by four positions**.

Ciphertext

QIIX EX RSSR

Step-by-Step Decryption

Cipher Letter	Shift -4	Plain Letter
Q	M	M
I	E	E
I	E	E
X	T	T
Space	Space	Space
E	A	A
X	T	T
Space	Space	Space
R	N	N
S	O	O
S	O	O
R	N	N

Recovered Message

MEET AT NOON

Decryption Process

Cipher Text

Q I I X E X R S S R

↓

Shift every letter backward by 4

Q→M

I→E

$I \rightarrow E$

$X \rightarrow T$

$E \rightarrow A$

$X \rightarrow T$

$R \rightarrow N$

$S \rightarrow O$

$S \rightarrow O$

$R \rightarrow N$

↓

Original Message :

MEET AT NOON

Decryption Formula

$$P = (C - K) \bmod 26$$

Where:

- **P** = Plaintext
- **C** = Ciphertext
- **K** = Shift key (4)

Example

Cipher letter **Q**

$$Q = 16$$

$$16 - 4 = 12$$

$$12 = M$$

Therefore,

$$Q \rightarrow M$$

Advantages of Caesar Cipher

- Very simple and easy to implement.
- Requires minimal computational resources.
- Useful for learning basic cryptographic concepts.

- Fast encryption and decryption.
- Suitable for educational demonstrations and simple message obfuscation.

Limitations

- Provides very weak security because there are only **25 possible shift keys**.
- Easily broken using brute-force attacks by trying every shift.
- Vulnerable to frequency analysis since letter patterns remain similar.
- Not suitable for protecting sensitive or confidential information in modern communication.

Real-World Applications

Although the Caesar Cipher is not secure enough for modern use, it is still valuable in educational and introductory contexts:

- Teaching basic encryption and decryption concepts in cryptography courses.
- Demonstrating substitution cipher techniques.
- Creating simple puzzles and escape-room challenges.
- Obfuscating non-sensitive text for learning purposes.

Modern communication systems instead use advanced encryption algorithms such as **AES (Advanced Encryption Standard)**, **RSA**, and **ECC**, which provide significantly stronger security.

Conclusion

In this scenario, the student organization encrypted the message "MEET AT NOON" using a **Caesar Cipher with a shift of 4**, producing the ciphertext "QIIX EX RSSR". An authorized member who knows the shift value can decrypt the ciphertext by shifting each letter **backward by four positions**, successfully recovering the original message. While the Caesar Cipher is an excellent tool for understanding the fundamentals of cryptography, its limited key space makes it unsuitable for securing real-world confidential communications.

2. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.

b. Explain how the repeating keyword influences each character of the ciphertext.

Solution :

The Vigenère Cipher (Keyword-based Polyalphabetic Substitution)

The scenario describes the **Vigenère Cipher**, a polyalphabetic substitution cipher that improves upon simple shift ciphers by using a **repeating keyword** to vary the shift applied to each letter. Unlike the Caesar cipher (which uses one fixed shift for the whole message), the Vigenère cipher uses a *different* shift for each letter, determined by the corresponding letter of the keyword.

Encryption formula:

$$C_i = (P_i + K_i) \bmod 26$$

Decryption formula:

$$P_i = (C_i - K_i) \bmod 26$$

Where:

- P_i = numeric position of the i-th plaintext letter (A=0, B=1, ..., Z=25)
- K_i = numeric position of the corresponding keyword letter (keyword repeats to match plaintext length)
- C_i = numeric position of the resulting ciphertext letter

a. Encrypting "HELLO" with keyword "KEY" — Step by Step

Step 1: Repeat the keyword to match the length of the plaintext

Plaintext "HELLO" has 5 letters, so the keyword "KEY" (3 letters) is repeated cyclically:

Plaintext H E L L O

Keyword K E Y K E

Step 2: Convert each letter to its numeric position

Letter H E L L O

P (pos) 7 4 11 11 14

Key Letter K E Y K E

K (pos) 10 4 24 10 4

Step 3: Apply $C_i = (P_i + K_i) \bmod 26$ for each letter

Position	P	K	P + K	mod 26	Ciphertext
1 (H)	7	10	17	17	R
2 (E)	4	4	8	8	I
3 (L)	11	24	35	9	J
4 (L)	11	10	21	21	V
5 (O)	14	4	18	18	S

Encrypted Message: RIJVS

(b) Explain How the Repeating Keyword Influences Each Character of the Ciphertext

The **repeating keyword** is the key feature that makes the Vigenère Cipher more secure than a simple Caesar Cipher.

Explanation

- The keyword **KEY** has only three letters.
- Since the plaintext contains five letters, the keyword repeats as **KEYKE**.
- Each letter of the keyword provides a different shift value:
 - **K** shifts the corresponding plaintext letter by **10** positions.
 - **E** shifts it by **4** positions.
 - **Y** shifts it by **24** positions.
- Because the shift changes with each character according to the repeated keyword, identical plaintext letters can encrypt to different ciphertext letters if paired with different keyword letters. For example, the two **L** characters in "HELLO" become **J** and **V** because they are encrypted using **Y** and **K**, respectively.

Influence Table

Plain Letter	Keyword Letter	Shift Value	Cipher Letter
H	K	10	R
E	E	4	I
L	Y	24	J
L	K	10	V
O	E	4	S

This varying shift pattern increases security by reducing predictable letter patterns in the ciphertext, making frequency analysis more difficult than with a Caesar Cipher.

Advantages of the Vigenère Cipher

- More secure than the Caesar Cipher because it uses multiple shifts.
- Repeating keyword creates varying encryption patterns.
- Reduces the effectiveness of simple frequency analysis.
- Easy to implement and understand for educational purposes.
- Demonstrates the concept of polyalphabetic substitution.

Limitations

- If the keyword is short or reused frequently, attackers may discover it using methods such as the Kasiski examination or Index of Coincidence.
- Not secure enough for modern applications due to advances in cryptanalysis.

- Modern systems use stronger encryption algorithms such as **AES**, **RSA**, and **ECC** for secure communication.

Conclusion

In this scenario, the company uses the **Vigenère Cipher** to encrypt the message "**HELLO**" with the keyword "**KEY**". By repeating the keyword as **KEYKE** and applying a different shift to each character, the plaintext is transformed into the ciphertext "**RIJVS**". The repeating keyword causes each character to be encrypted using a different shift value, providing greater security than a fixed-shift cipher such as the Caesar Cipher. Although the Vigenère Cipher is no longer considered secure for real-world communication, it remains an important classical encryption technique for understanding the principles of cryptography.